



(An Autonomous Institute Affiliated to VTU, Belagavi, Accredited by NAAC with 'A' Grade)

Department of Robotics & Artificial Intelligence

Duration: 90 Min

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DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute Affiliated to VTU, Belagavi, Accredited by NAAC with 'A' Grade)

Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560111

Department of Robotics & Artificial Intelligence

Continuous Internal Assessment Test-III

Course: **Fundamentals of Robotics and Applications**

Maximum marks: 50

Course Code: **22RI32**

Date: 2-12-2024

Semester: III

Duration: 90 Min

Question No.	1	2	3	4	4	5	6	6	7	8	8
	a	a	a	a	b	a	a	b	a	a	b
Course Outcome	4	4	4	5	5	5	4	4	5	5	5
Program Outcome	2	2	2	2	2	3	2	2	2	2	2
Blooms Level	2	2	2	2	2	3	2	3	2	2	3

Answer five full questions, questions 1 and 2 are compulsory all questions carry equal marks		Marks
1	What are the ways of accomplishing lead through programming? Explain detail manual lead through programming method in robot application.	10
2	List the components of Optical encoder and with a neat labelled diagram of Incremental encoder mention its salient features.	10
3	Describe method of Robot programming with textual languages. Explain the First-generation languages.	10
(OR)		
4 a)	Define sensor. Give classification of sensors with their functions.	5
b)	With a neat sketch explain the Robot language structure.	5
5	Illustrate machine vision system concept used in industry.	10
(OR)		
6 a)	Write the neat labelled diagram of Potentiometer and illustrate its working process	5
b)	Illustrate velocity sensors with neat diagram.	5
7	What are the General consideration in trajectory planning? Explain	10
(OR)		
8 a)	Discuss different Schemes of path generation with advantages and disadvantages.	6
b)	Differentiate path planning, motion planning and trajectory planning.	4

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Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560111

UG Semester End Examination, January/February - 2025

Course: **Fundamentals of Robotics and Applications** Maximum marks: **100**
 Course Code: **22RI32** Duration: **03 hours**
 Semester: **III**

Note: i) Answer any FIVE full questions. ii) Questions 1 and 2 are compulsory.
 iii) Any missing data should be suitably assumed. iv) Verify the course code before answering.

Q. No.		Marks
1	a) Define automation. With the help of a neat sketch explain the three main categories of industrial automation and explain each briefly. 08 b) What are the three laws of robotics formulated by Issac Asimov? 06 c) What are the various specifications of a robot? List them and provide a detailed explanation of any one. 06	
2	a) You are required to design a robot that can pick up an object of arbitrary shape from a table and place it inside a box, 10 i) What is the minimum number of degrees of freedom it should have? ii) List all the possible arm configurations. iii) Draw their corresponding work volumes. b) Determine the homogeneous transformation matrix to represent the following sequence of operations. 10 i) Rotation of 60° about OX axis ii) Translation of 4 units along OX axis iii) Translation of -6 units along OC axis iv) Rotation of 30° OB axis	
3	a) Discriminate the salient features, capabilities, applications, merits and limitations of non-servo and servo controlled robots. 10 b) Explain vacuum grippers, with reference to the principle and applications. 10	
OR		
4	a) Discriminate the factors which must be considered while choosing the drive system for robots. 10 b) Explain briefly point to point contact and continuous path control robot. 10	
5	a) Classify the different types of programming methods and illustrate any one in detail 10 b) With neat sketch explain the following: i) Potentiometer ii) Optical Incremental Encoders. 10	
OR		
6	a) What are the ways of accomplishing lead through programming? Explain detail manual lead through programming method in robot application. 10 b) What are the three function of vision system explain and also mention applications. 10	
7	a) Enumerate trajectory generation polynomial types. 10 b) Illustrate the parameters involved in the path planning with 3 rd degree polynomial. 10	
OR		
8	a) Evaluate the advantages of robotic welding and assembly systems in manufacturing industries, considering factors such as precision, speed, and quality control. 10 b) Explore the applications of robotics in agriculture, such as precision farming, crop monitoring, and harvesting, highlighting their potential to increase yields and reduce resource consumption. 10	